

Perovskite windows to reduce energy load in buildings

According to a research effort led by the U.S. Department of Energy's National Renewable Energy Laboratory (NREL), thermochromic windows in office buildings improve energy efficiency across all climate zones in the United States by modulating the temperature inside, leading to a massive

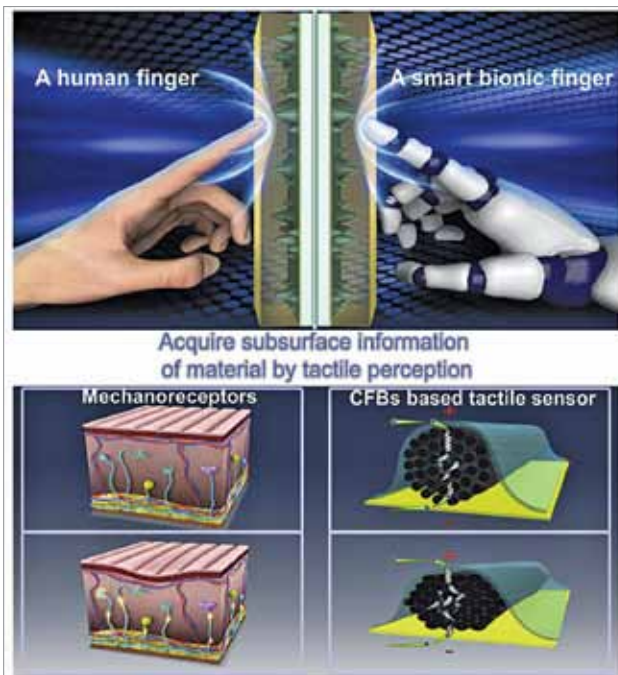


Downtown Denver features many buildings with glass facades. Researchers from NREL say retrofitting these windows with thermochromic ones can improve energy efficiency across all climate zones in the US (Credit: Dennis Schroeder, NREL)

savings. Thermochromic windows are based on the use of perovskite materials to absorb energy from the sun shift from transparent to a visibly absorbing or reflecting state. Researchers developed a model in which they built a 12-story structure with a window-to-wall ratio of 95%. They simulated energy use in the building in 15-minute intervals for a year in eight climate zones across the country, covering Hawaii, Arizona, California, Colorado, New York, Wisconsin, Minnesota, and Alaska. They determined that thermochromic double-pane windows improved building energy efficiency over double-pane windows in each zone and energy savings were greater in colder regions as well, as the thermochromic double-pane windows outperformed even triple-pane windows in the hottest climate zones.

Bionic finger creates 3D internal maps of objects by touching their surface

Researchers from Wuyi University have developed a bionic finger that can create 3D maps of the internal shapes and textures of complex objects by touching their exterior surface. The inspiration for this device comes from the human finger with which we can sense not only the texture of our skin but also the outline of the bone beneath it. The bionic finger "scans" an object by moving across it and applying pressure—think of a constant stream of pokes or prods. With each poke, the carbon fibres compress, and the



Graphical abstract (Credit: Cell Reports Physical Science, 2023)

degree to which they compress provides information about the relative stiffness or softness of the object. Depending on the object's material, it might also compress when poked by the bionic finger: rigid objects hold their shape, while soft objects deform when enough pressure is applied. This information, along with the location at which it was recorded, is relayed to a personal computer and displayed on screen as a 3D map.

Smart contact lenses that can implement AR based navigation

Researchers from Ulsan National Institute of Science and Technology (UNIST) and KERI have developed core technology for smart contact lenses that can implement augmented reality (AR) based navigation, with a 3D printing process. This technology can realise AR by printing micro-patterns on a lens display using a 3D printer without applying voltage. The key is the meniscus of used ink. The meniscus of the acidic-ferrocyanide ink is formed on the substrate when the ink-filled micronozzle and substrate come in contact. When water evaporates from the meniscus, the water molecules and precursor ions move toward the meniscus surface by convective flow, generating a preferential accumulation of the precursor ions in the outer part of the meniscus. Simply by wearing a lens, navigation unfolds in front of a person's eyes through AR. Games such as the popular "Pokemon Go" can also be enjoyed with smart contact lenses, not just smartphones.